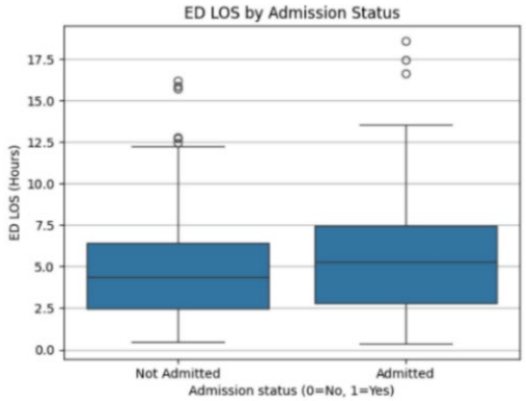
SkillUp

ED Data Analytics Project

Task Number	Task name	Code text/screenshot	Output text/screenshot

1	**Inspect dataset dimensions and column types**	`#Basic Inspection`
		`print('Rows, Columns:', df.shape)`
		`print('\nColumn Data Types:')`
		`print(df.dtypes)`
			`Rows, Columns: (400, 7)`
			
			`Column Data Types:`
			`visit_id int64`
			`age int64`
			`triage_level object`
			`arrival_hr int64`
			`ed_los_hours float64`
			`num_tests_ordered int64`
			`admitted int64`
			`dtype: object`

2	**Summarize ED length of stay**	`#LOS mean and median`
		`print('Mean LOS in ED:', df['ed_los_hours'].mean())`
		`print('Median LOS in ED:', df['ed_los_hours'].median())`
			`Mean LOS in ED: 5.0272`
			`Median LOS in ED: 4.535`

3	**Compare LOS by admission status**	`#Comparing ED LOS (Hours) for Admitted vs. Non-Admitted Patients (Box-and-Whisker Plot)`
		`import seaborn as sns`
		`import matplotlib.pyplot as plt`
		
		`sns.boxplot(x = 'admitted', y = 'ed_los_hours', data =df)`
		`plt.title('ED LOS by Admission Status')`
		`plt.xlabel('Admission status (0=No, 1=Yes)')`
		`plt.xticks(ticks=[0, 1], labels=['Not Admitted', 'Admitted'])`
		`plt.ylabel('ED LOS (Hours)')`
		`plt.grid(axis='y')`
		`plt.show()`
			

4	**Hypothesis test: Admission vs LOS**	`#Hypothesis Test`
		`from scipy import stats`
		
		`# Separate into admitted vs. non-admitted groups for comparison:`
		`los_admitted = df[df['admitted'] ==1]['ed_los_hours']`
		`los_nonadmitted =df[df['admitted'] ==0]['ed_los_hours']`
		
		`# t-test (independent)`
		`t_statistic, p_value = stats.ttest_ind(los_admitted, los_non_admitted)`
		`print(f"t-statistic: {t_statistic:.3f}")`
		`print(f"p-value: {p_value:.3f}")`
		
		`if p_value < 0.05:`
		`  print("The mean ED LOS is statistically significantly different between admitted and non-admitted patients (p < 0.05).")`
		`else:`
		`  print("There is no statistically significant difference in mean ED LOS between admitted and non-admitted patients (p >= 0.05).")`
			`t-statistic: 2.070`
			`p-value: 0.039`
			`The mean ED LOS is statistically significantly different between`
			`admitted and non-admitted patients (p < 0.05).`

5	**ANOVA: LOS across triage levels**	`#LOS across Triage Levels: One-Way ANOVA Test`
		`from scipy.stats import f_oneway`
		
		`#Separate LOS by triage level`
		`los_low = df[df['triage_level']=='Low']['ed_los_hours']`
		`los_medium = df[df['triage_level']=='Medium']['ed_los_hours']`
		`los_high = df[df['triage_level']=='High']['ed_los_hours']`
		
		`#One-Way ANOVA`
		`f_statistic, p_value = f_oneway(los_low, los_medium, los_high)`
		`print('One-Way ANOVA:')`
		`print(f"F-statistic: {f_statistic:.3f}")`
		`print(f"p-value: {p_value:.3f}")`
			`One-Way ANOVA:`
			`F-statistic: 0.121`
			`p-value: 0.886`

6	**Logistic regression: Admission risk**	`#Admission Risk Using Logistic Regression`
		`from sklearn.linear_model import LogisticRegression`
		`X = df[['age', 'ed_los_hours', 'num_tests_ordered']]`
		`y = df['admitted']`
		
		`logistic_model = LogisticRegression()`
		`logistic_model.fit(X,y)`
		`print(f"Model Accuracy: {logistic_model.score(X, y):.3f}")`
			`Model Accuracy: 0.695`

7	**Evaluate admission model (AUC)**	`#Evaluate Admission Model (AUC)`
		`from sklearn.metrics import roc_auc_score`
		
		`y_pred_proba = Logistic_model.predict_proba(x)[:,1]`
		`auc_score = roc_auc_score(y, y_pred_proba)`
		`print(f"AUC Score: {auc_score:.3f}")`
			`AUC Score: 0.560`

8	**Decision tree model**	`#Training a Decision Tree Model`
		`from sklearn.tree import DecisionTreeClassifier`
		
		`decision_tree_model = DecisionTreeClassifier()`
		`decision_tree_model.fit(X,y)`
		`print(f"Model Accuracy: {decision_tree_model.score(X,y):.3f}")`
			`Model Accuracy: 1.000`

9	**Random forest comparison**	`#Random Forest Comparison`
		`from sklearn.ensemble import RandomForestClassifier`
		
		`random_forest_model = RandomForestClassifier(random_state=42)`
		`random_forest_model.fit(X,y)`
		`print(f"Model Accuracy: {random_forest_model.score(X,y):.3f}")`
			`Model Accuracy: 1.000`

Operational Interpretation: Patients with longer emergency department length of stay and a higher number of diagnostic tests have an increased likelihood of hospital admission, suggesting these variables can be used for early risk stratification.

Key Statistical Analysis Insights: The t-test of independent samples (or unpaired t-test) resulted in a p-value of < 0.05, indicating a statistically significant difference between admitted and non-admitted patients. A high logistic regression accuracy of 0.69 for hospital readmission and a low AUC (marginally better than random chance) could be due to class imbalance. The one-way ANOVA test showed no significant difference between admitted and non-admitted groups based on low, medium or high triage level. To improve the model, 1) add other clinically significant variables and 2) use regularization (eg LASSO) to enhance the model's generalizability and performance on new data by preventing overfitting.